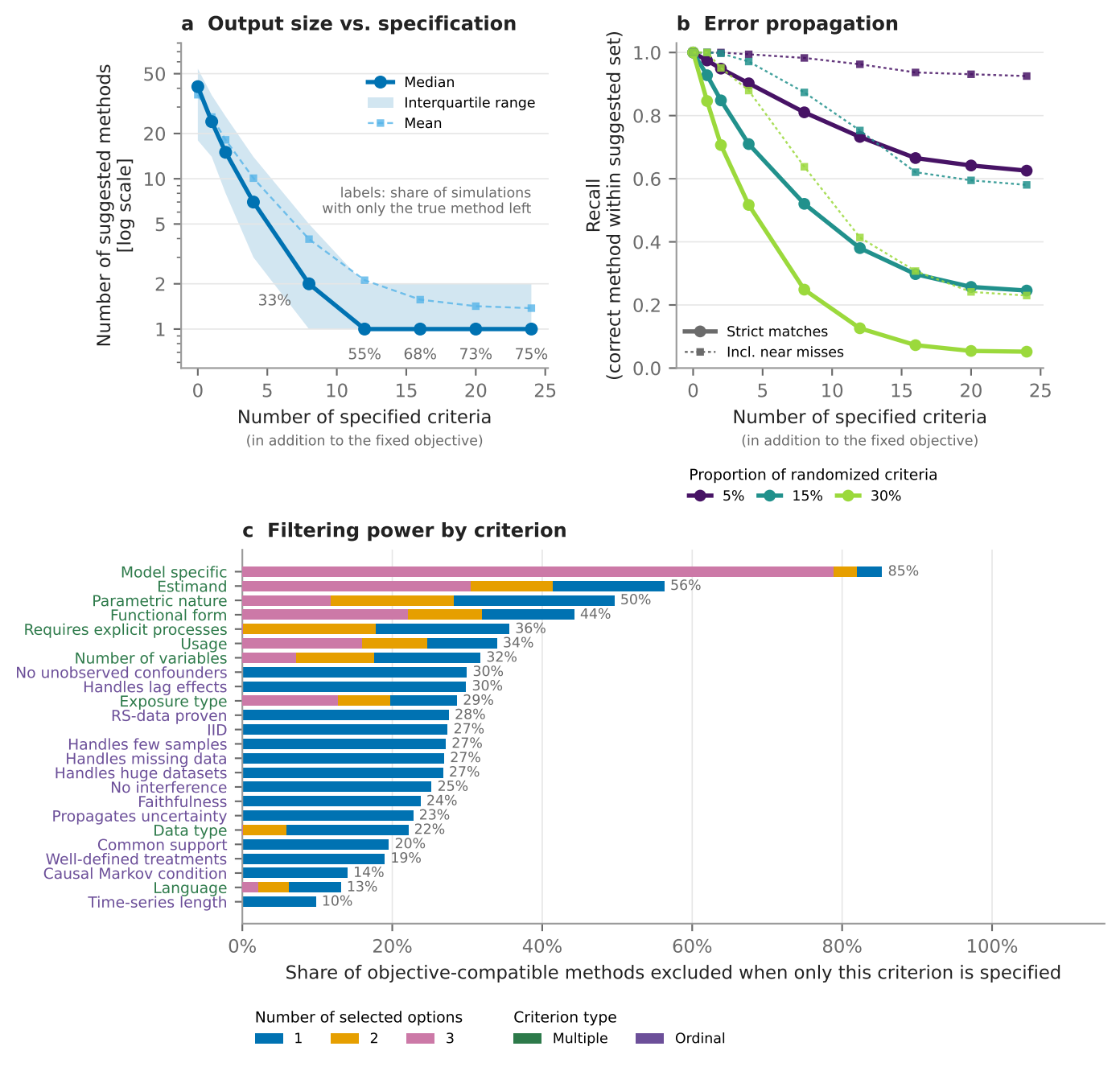
a Output size vs. specification
Number of suggested methods [log scale]
50
20
10
5
2
1
Median
Interquartile range
Mean
labels: share of simulations with only the true method left
33%
55%
68%
73%
75%
0
5
10
15
20
25
Number of specified criteria
(in addition to the fixed objective)
b Error propagation
Recall (correct method within suggested set)
1.0
0.8
0.6
0.4
0.2
0.0
Strict matches
Incl. near misses
0
5
10
15
20
25
Number of specified criteria
(in addition to the fixed objective)
Proportion of randomized criteria
5%
15%
30%
c Filtering power by criterion
Model specific 85%
Estimand 56%
Parametric nature 50%
Functional form 44%
Requires explicit processes 36%
Usage 34%
Number of variables 32%
No unobserved confounders 30%
Handles lag effects 30%
Exposure type 29%
RS-data proven 28%
IID 27%
Handles few samples 27%
Handles missing data 27%
Handles huge datasets 27%
No interference 25%
Faithfulness 24%
Propagates uncertainty 23%
Data type 22%
Common support 20%
Well-defined treatments 19%
Causal Markov condition 14%
Language 13%
Time-series length 10%
0%
20%
40%
60%
80%
100%
Share of objective-compatible methods excluded when only this criterion is specified
Number of selected options
1
2
3
Criterion type
Multiple
Ordinal